

Project -02 | Application Trend

Step 01 – Clone the repo to local EC2 Ubuntu machine

git clone <https://github.com/Vennilavanguvi/Trend.git>

Version Control

Create a repo on git and have pushed all the codes

<https://github.com/singh-ajmer-git/trend.git>

Branches

Infra - for infra codes, including ECR cluster & monitoring

Dev - application & Docker file

Monitoring - Prometheus and monitoring

Branches New branch

Overview Yours Active Stale All

Q Search branches...

Default

Branch	Updated	Check status	Behind	Ahead	Pull request
infra	3 hours ago			Default	...

Your branches

Branch	Updated	Check status	Behind	Ahead	Pull request
monitoring	now		2	1	...
dev	1 hour ago		2	8	...

Active branches

Branch	Updated	Check status	Behind	Ahead	Pull request
monitoring	now		2	1	...
dev	1 hour ago		2	8	...

Docker– Write the Docker file to build an image.

Vi Dockerfile

root@ip-172-31-35-67:/home/ubuntu/Trend# cat Dockerfile

```
FROM public.ecr.aws/nginx/nginx:alpine
```

```
COPY dist/ /usr/share/nginx/html
```

EXPOSE 80

CMD ["nginx", "-g", "daemon off;"]

Build the image locally to test it.

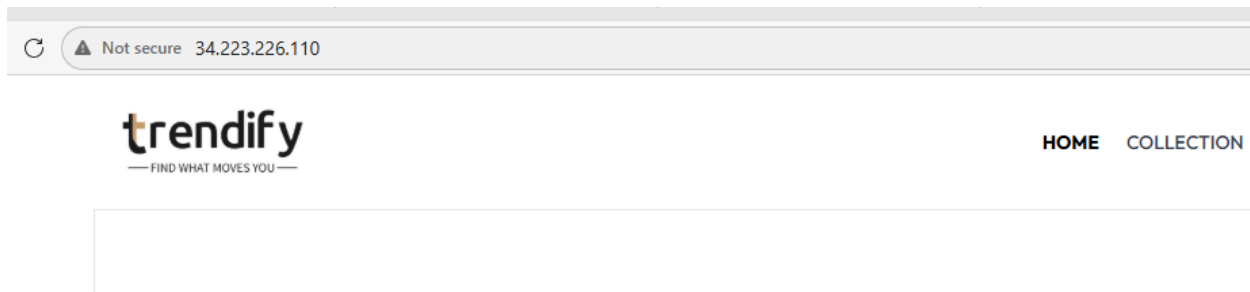
docker build -t trend-aj:v3

```
root@ip-172-31-30-99:/home/ubuntu/trend/dev# ls
Dockerfile  deployment.yaml  dist  service.yaml
root@ip-172-31-30-99:/home/ubuntu/trend/dev# cat Dockerfile
FROM public.ecr.aws/nginx/nginx:alpine
COPY dist/ /usr/share/nginx/html
EXPOSE 80
CMD ["nginx", "-g", "daemon off;"]
root@ip-172-31-30-99:/home/ubuntu/trend/dev# docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
524781515606.dkr.ecr.us-west-2.amazonaws.com/trend-repo:v3	751796a71ea5	111MB	35MB	U
build-frontend:latest	3373527c33f9	96MB	26.7MB	
hello-world:latest	85404b3c5395	25.9kB	9.52kB	
jenkins/jenkins:lts	23b84d1066db	802MB	288MB	
prom/alertmanager:latest	88b605de9aba	119MB	39.1MB	U
prom/blackbox-exporter:latest	e753ff9f3fc4	57MB	17.6MB	U
prom/prometheus:latest	4a61322ac110	535MB	144MB	U
react-app:latest	ccf3abb614f9	98.2MB	27.6MB	
singajmihub/react-app-dev:latest	ae6daad5a731	98.6MB	27.7MB	
singajmihub/react-app-prod:latest	fdff60108fa9	98.6MB	27.7MB	
singajmihub/react-app:latest	d92baa685f52	98.2MB	27.6MB	
trend:v1	f055c9fa5220	111MB	35MB	U
trend:v3	751796a71ea5	111MB	35MB	U

```
root@ip-172-31-30-99:/home/ubuntu/trend/dev# docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS                               NAMES
62b37d62ba11  751796a71ea5  "/docker-entrypoint...."  36 seconds ago Up 36 seconds  0.0.0.0:80->80/tcp, [::]:80->80/tcp  trendtestcon
root@ip-172-31-30-99:/home/ubuntu/trend/dev#
```

Run a test container to validate the image:



Terraform-

Setup the communication, authentication & authorization to AWS from ubuntu machine.

```
root@ip-172-31-35-67:/home/ubuntu/Trend# aws configure list
NAME      : VALUE                                : TYPE      : LOCATION
profile   : <not set>                             : None      : None
access_key : *****4Z2D                          : iam-role   :
secret_key : *****0j0t                          : iam-role   :
region    : us-west-2                             : imds       :
root@ip-172-31-35-67:/home/ubuntu/Trend#
```

This enables to push the terraform code from local to AWS API.

Code File – main.tf (containers VPC, IAM, EC2), provider.tf setup the code.

Run terraform CMD to provision the infra

terraform init

terraform plan

Terraform apply –auto-approve

```
root@ip-172-31-30-99:/home/ubuntu/trend/infra# terraform apply -auto-approve
data.aws_security_group.existing_sg: Reading...
data.aws_vpc.existing_vpc: Reading...
data.aws_security_group.existing_sg: Read complete after 0s [id=sg-0f17f39f935d2e732]
data.aws_vpc.existing_vpc: Read complete after 0s [id=vpc-020409da3e274321b]
data.aws_subnet.existing_subnet: Reading...
data.aws_subnet.existing_subnet: Read complete after 0s [id=subnet-05fd91c15272d41ff]
```

```
Apply complete! Resources: 3 added, 0 changed, 0 destroyed.
root@ip-172-31-30-99:/home/ubuntu/trend/infra# terraform state list
data.aws_security_group.existing_sg
data.aws_subnet.existing_subnet
data.aws_vpc.existing_vpc
aws_iam_instance_profile.ec2_profile
aws_iam_role.ec2_role
aws_instance.web
root@ip-172-31-30-99:/home/ubuntu/trend/infra#
```

Added all codes to respective git repo. Initiated, committed, and pushed the code to repo.
This includes giving permission to the repo in PAT token.

```
root@ip-172-31-35-67:/home/ubuntu/trend# terraform state list
aws_iam_instance_profile.ec2_profile
aws_iam_role.ec2_role
aws_instance.web
aws_subnet.trend_public_subnet
aws_vpc.trend_vpc
root@ip-172-31-35-67:/home/ubuntu/Trend#
```

Instances (1/2) Info

Last updated less than a minute ago Refresh Connect Insta

Find Instance by attribute or tag (case-sensitive)

All states

☐	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone
☒	lab	i-0f8829451ba6dc51c	Running	t3.large	3/3 checks passed	View alarms +	us-west-2b
☐	trend-ec2-jenkins	i-0a373ff19606d2263	Running	t2.micro	2/2 checks passed	View alarms +	us-west-2a

Your VPCs

VPCs VPC encryption controls

Your VPCs (1) Info

Last updated less than a minute ago

Find VPCs by attribute or tag

VPC ID: vpc-020409da3e274321b Clear filters

☐	Name	VPC ID	State	Encryption c...	Encryption control ...	Block Public...	IPv4 CIDR
☐	VPN-lb	vpc-020409da3e274321b	Available	-	-	Off	172.31.0.0/16

trend-ec2-role Info

Summary

Creation date

March 24, 2026, 03:36 (UTC+05:30)

ARN

arn:aws:iam::524781515606:role/trend-ec2-role

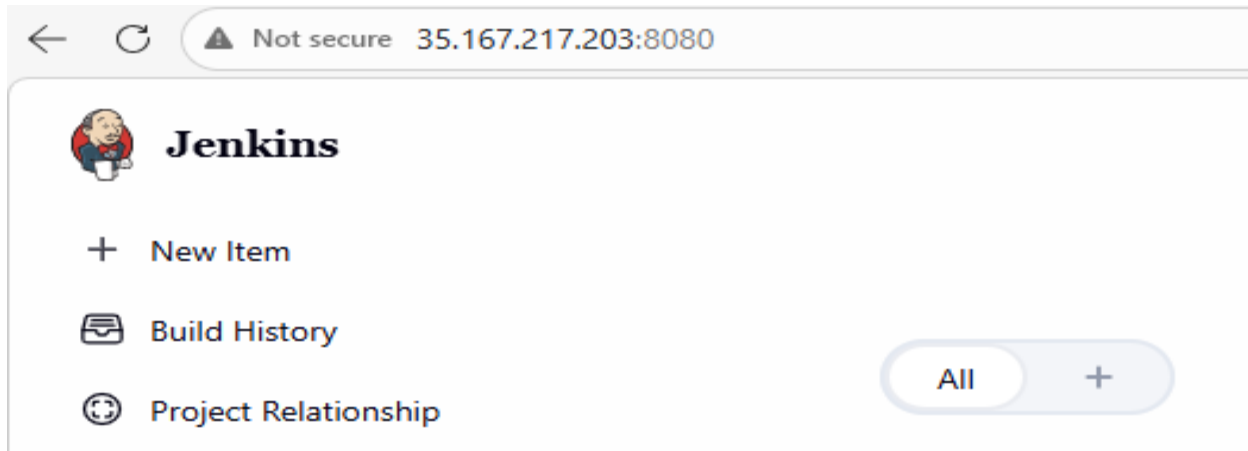
Last activity

Maximum session duration

Jenkins provisioned

```
root@ip-172-31-30-99:/home/ubuntu/trend/infra# systemctl status jenkins
● jenkins.service - Jenkins Continuous Integration Server
   Loaded: loaded (/usr/lib/systemd/system/jenkins.service; enabled; preset: enabled)
   Drop-In: /etc/systemd/system/jenkins.service.d
            └─override.conf
   Active: active (running) since Tue 2026-03-24 20:16:56 UTC; 23min ago
   Main PID: 549 (java)
     Tasks: 43 (limit: 9301)
    Memory: 592.5M (peak: 597.0M)
       CPU: 30.608s
    CGroup: /system.slice/jenkins.service
            └─549 /usr/bin/java -Xms1g -Xmx4g -XX:+UseG1GC -Djava.awt.headless=true -jar /usr

Mar 24 20:16:52 ip-172-31-30-99 java[549]: 2026-03-24 20:16:52.615+0000 [id=39] INFO
Mar 24 20:16:52 ip-172-31-30-99 java[549]: 2026-03-24 20:16:52.625+0000 [id=38] INFO
Mar 24 20:16:53 ip-172-31-30-99 java[549]: 2026-03-24 20:16:53.383+0000 [id=37] INFO
Mar 24 20:16:55 ip-172-31-30-99 java[549]: 2026-03-24 20:16:55.694+0000 [id=38] INFO
```



Kubernetes

Deployed EKS cluster via cluster.yaml

```
eksctl create cluster -f cluster.yaml
```

```
aws eks update-kubeconfig --region us-west-2 --name trend-eks
```

Cluster is running fine on AWS

> Clusters

Clusters (1) [Info](#)

🔍 Filter clusters

Cluster name	Status	Kubernetes version	Support period	Upgrade policy
trend-eks	Active	1.34 Upgrade now	Standard support until December 2, 2026	Extended support

```
root@ip-172-31-30-99:/home/ubuntu/trend/infra# kubectl get nodes
NAME                                STATUS    ROLES    AGE    VERSION
ip-172-31-23-202.us-west-2.compute.internal Ready    <none>   3m25s  v1.34.4-eks-f69f56f
ip-172-31-41-120.us-west-2.compute.internal Ready    <none>   3m24s  v1.34.4-eks-f69f56f
root@ip-172-31-30-99:/home/ubuntu/trend/infra#
```

Wrote Deployment & Service yaml files

Setup the Jenkins pipeline to pull the updated code from git repo and deploy it on cluster full CI/CD

The Jenkinsfile read the repo and files deployment & service.yaml and push the deployment on cluster trend-eks

Jenkins pipeline

35.167.217.203:8080/job/trend-app-pipeline/

Jenkins / trend-app-pipeline via Jen...

Status

Configure

Scan Multibranch Pipeline Now

Scan Multibranch Pipeline Log

Multibranch Pipeline Events

Delete Multibranch Pipeline

Build History

Next Steps

trend-app-pipeline via Jenkinsfile

Folder name: trend-app-pipeline
Trend app deployment via jenkins pipeline

Branches (1)

S	W	Name ↓	Last Success	Last Failure
✓	☀	dev	1 hr 26 min #2	N/A

Enabled ☒

Display Name ?
trend-app-pipeline via Jenkinsfile

Description
Trend app deployment via jenkins pipeline
Plain text [Preview](#)

Branch Sources

Git

Project Repository ?
https://github.com/singh-ajmer-git/trend.git

Credentials ?
- none - [+ Add](#)

This trigger based on Webhooks at Git>> followed the Jenkinsfile and deployed the resources on EKS.

Webhooks

Add webhook

Webhooks allow external services to be notified when certain events happen. When the specified events happen, we'll send a POST request to each of the URLs you provide. Learn more in our [Webhooks Guide](#).

✓ http://35.167.217.203:8080/github-... (push)

Edit Delete

Last delivery was successful.



Jenkins

trend-app-pipeline via Jen...

Branches (1)

Build History



Status



Configure



Scan Multibranch Pipeline Now



Scan Multibranch Pipeline Log



Multibranch Pipeline Events



Build History

Build History of Branches (1)

S	Build	
✓	trend-app-pipeline via Jenkinsfile » dev	#2
✓	trend-app-pipeline via Jenkinsfile » dev	#1

```
root@ip-172-31-30-99:/home/ubuntu/trend/dev# kubectl get all -n monitoring
NAME                                     READY   STATUS    RESTARTS   AGE
pod/alertmanager-76c6d9bb7c-6g2nm       1/1     Running   0           58m
pod/blackbox-exporter-6dfff7fd7c-1jxr4  1/1     Running   0           59m
pod/prometheus-667975857f-g27pw         1/1     Running   0           58m

NAME                                     TYPE          CLUSTER-IP      EXTERNAL-IP
service/alertmanager                    ClusterIP      10.100.199.133   <none>
service/blackbox-exporter                ClusterIP      10.100.28.155    <none>
service/prometheus                       LoadBalancer  10.100.174.209   af2fb5eaa6e2405e8c1e9c877e04f2e-2108214246.us-west-2.elb.amazonaws.com

NAME                                     READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/alertmanager             1/1     1             1           58m
deployment.apps/blackbox-exporter         1/1     1             1           59m
deployment.apps/prometheus                1/1     1             1           58m

NAME                                     DESIRED   CURRENT   READY   AGE
replicaset.apps/alertmanager-76c6d9bb7c  1         1         1       58m
replicaset.apps/blackbox-exporter-6dfff7fd7c  1         1         1       59m
replicaset.apps/prometheus-667975857f       1         1         1       58m
root@ip-172-31-30-99:/home/ubuntu/trend/dev#

root@ip-172-31-30-99:/home/ubuntu/trend/dev# kubectl get svc
NAME                                     TYPE          CLUSTER-IP      EXTERNAL-IP      PORT(S)
kubernetes                             ClusterIP      10.100.0.1       <none>            443/TCP
trend-app-service                       LoadBalancer  10.100.189.174   aa5b56bf7b561415eaaeb62116b56f38-2079953017.us-west-2.elb.amazonaws.com  80:30549
TCP 98m

root@ip-172-31-30-99:/home/ubuntu/trend/dev# kubectl get deployment
NAME      READY   UP-TO-DATE   AVAILABLE   AGE
trend-app 2/2     2             2           111m
root@ip-172-31-30-99:/home/ubuntu/trend/dev#
```

Monitoring-

Deployed the Prometheus to monitor the Cluster:

← Not secure af2fb5eaf6e2405e8c1e9c877e04f2e-2108214246.us-west-2.elb.amazonaws.com:9090/targets

Prometheus Query Alerts Status Target health

Select scrape pool Filter by target health Filter by endpoint or labels

Hiding pools with no matching targets
Hiding 1 empty pools due to filters or no targets. Show empty pools

blackbox 1 / 1 up

Endpoint	Labels	Last scrape	State
http://blackbox-exporter.monitoring.svc.cluster.local:9115/probe module="http_3xx" target="http://x1c1fe74f62074640bba12ca3c177c07-1072782686.us-west-2.elb.amazonaws.com:80"	instance="http://x1c1fe74f62074640bba12ca3c177c07-1072782686.us-west-2.elb.amazonaws.com:80" job="blackbox"	1.817s ago 7ms	UP

Finally, the app is running in container format

Not secure aa5b56bf7b561415eaaeb62116b56f38-2079953017.us-west-2.elb.amazonaws.com

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